

LLM Benchmarking for Emergency Triage

April 23, 2026

1 Introduction

Emergency department (ED) triage is a high-stakes clinical process because it determines how scarce medical resources are prioritised under time pressure. Triage errors can create two different harms: *under-triage* may delay urgent treatment for truly high-risk patients, whereas *over-triage* may send low-risk patients into emergency pathways and worsen crowding. Standard ED triage systems such as the Emergency Severity Index (ESI) were developed to support structured prioritisation, yet both under- and over-triage remain clinically and ethically important problems [1].

Large language models (LLMs) are increasingly being explored as clinical decision-support tools, including for emergency triage. Existing evidence suggests that LLMs can generate plausible triage recommendations, but performance varies across models and settings. A recent systematic review and meta-analysis found that ChatGPT-4.0 outperformed ChatGPT-3.5 in ED triage tasks, but also reported substantial heterogeneity across studies, indicating that clinical readiness remains unresolved [2]. More recent large-scale retrospective evidence likewise suggests that current LLMs are better treated as adjunctive tools than autonomous triage systems [3].

This project addresses the issue through a small but controlled benchmark designed not only to assess overall triage performance, but also to probe demographic sensitivity. We ask three questions: (1) How safe and accurate are LLM triage recommendations? (2) Do demographic variables influence outputs in ways that raise fairness concerns? (3) What governance framework is needed for safe and accountable clinical use?

2 Methods

2.1 Benchmark Dataset

To support a controlled fairness evaluation, we constructed a benchmark file, `triage_vignettes_full.csv`, containing 40 rows in total: 10 base cases and 30 demographic variants. Each base case represents a realistic ED triage vignette, and each selected case was expanded into three variants by modifying only *age*, *sex*, and *insurance status* while keeping the clinical text unchanged. This design allows us to test whether a model changes its recommendation when demographic information changes but the underlying clinical presentation remains constant.

The dataset schema includes `case_id`, `version_id`, `text`, `gold_level`, `age`, `sex`, and `insurance`. The final urgency labels use a simplified three-level scheme: `em` (emergency), `ne` (non-emergency),

and *sc* (self-care). Gold labels were assigned using project-defined triage criteria aligned with standard ED urgency logic and then checked for internal consistency.

2.2 Model Evaluation Pipeline

We evaluated two models, *DeepSeek-Chat* and *Kimi (Moonshot)*, through API-based benchmarking. No model training or fine-tuning was performed. For each case, the pipeline constructed a prompt and asked the model to return a structured triage label and short reasoning. We tested two prompting conditions:

- **text_only**: the model received only the clinical vignette text
- **text_plus_demo**: the model received the vignette text plus explicit age, sex, and insurance information

This design allowed us to compare not only model performance, but also whether prompt structure itself altered fairness or consistency.

2.3 Evaluation Metrics

We used four main metrics:

1. **Accuracy**: proportion of model outputs matching the gold label
2. **Under-triage rate**: proportion of true *em* cases incorrectly downgraded to *ne* or *sc*
3. **Over-triage rate**: proportion of lower-acuity cases incorrectly escalated to *em*
4. **Demographic consistency**: whether the model produced stable outputs across the demographic variants of the same *case_id*

We also examined subgroup results by *sex* and *insurance status* to assess whether performance differed systematically across patient groups.

3 Results

3.1 Overall Performance

Table 1 summarises the main benchmark results.

Table 1: Overall benchmark results across models and prompting conditions.

Model	Condition	Accuracy	Under-triage	Over-triage	Consistency
DeepSeek	text_only	62.5%	0.0%	37.5%	90%
DeepSeek	text_plus_demo	72.5%	0.0%	27.5%	80%
Kimi	text_only	75.7%	0.0%	24.3%	100%
Kimi	text_plus_demo	71.4%	0.0%	28.6%	100%

Two patterns are immediately clear. First, no under-triage was observed in any model or condition, which is encouraging from a patient-safety perspective. Second, both models displayed moderate over-triage, suggesting a conservative tendency to assign higher urgency than necessary. DeepSeek improved substantially when demographic information was added to the prompt, whereas Kimi performed slightly worse under the same prompt change. This indicates that the effect of prompt enrichment is model-dependent rather than universally beneficial.

3.2 Fairness and Consistency Findings

The subgroup analysis revealed that fairness patterns differed by model. For DeepSeek under *text_only*, female cases showed lower accuracy than male cases (58.8% vs. 69.2%). Under *text_plus_demo*, female accuracy increased sharply to 82.4%, suggesting that explicit demographic prompting improved performance for this subgroup in DeepSeek. Kimi showed the opposite pattern: it maintained higher male accuracy in both conditions, and the female–male gap widened under *text_plus_demo* (64.3% vs. 81.8%). Insurance-related patterns were also heterogeneous. DeepSeek became more balanced under *text_plus_demo*, whereas Kimi showed lower accuracy for uninsured cases when demographic information was added. These results suggest that demographic prompting can either reduce or amplify subgroup disparities depending on the model.

Consistency was high overall, but again differed by model. Kimi achieved 100% consistency across demographic variants in both conditions, whereas DeepSeek ranged from 80% to 90%. This suggests that Kimi was more stable in its treatment of controlled patient variants, while DeepSeek was somewhat more responsive—or more unstable—when demographic cues were added. High consistency is desirable, but consistency alone should not be mistaken for fairness, because a model can be consistently wrong for an entire subgroup.

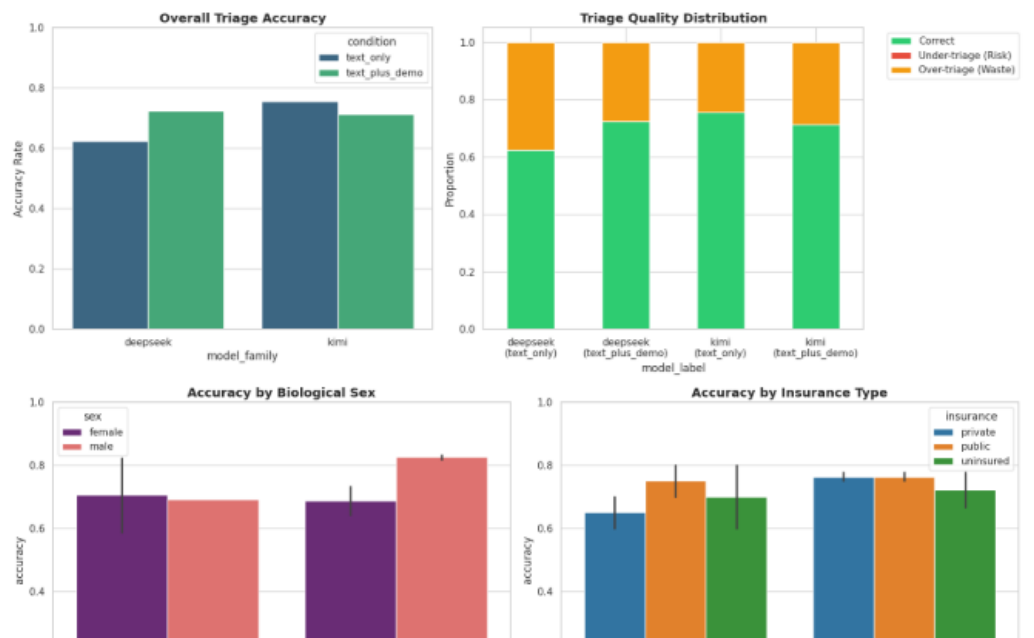


Figure 1: Benchmark summary visualisation.

4 Discussion and Governance

Our benchmark suggests that current LLMs can achieve a strong baseline safety profile against under-triage, but do so through a conservative strategy that produces non-trivial over-triage. In triage ethics, these harms are not symmetrical: under-triage is more directly tied to delayed care and patient harm, whereas over-triage primarily burdens resource allocation and system efficiency [1]. From this perspective, the absence of under-triage in our benchmark is encouraging, but it does not justify clinical deployment on its own, because persistent over-triage may worsen ED crowding and reduce trust in the tool.

Fairness findings were more complex. Because the benchmark held clinical text constant across demographic variants, differences in model output can reasonably be interpreted as demographic sensitivity rather than changes in medical content. However, the direction of that sensitivity was not uniform. DeepSeek in some cases benefited from explicit demographic prompting, whereas Kimi in some cases became less equitable under the same intervention. This means fairness is not a stable property of “the model” alone; it emerges from the interaction between model architecture, prompt design, and benchmark structure.

A further concern is *automation bias*: the tendency for clinicians to over-rely on algorithmic outputs that appear coherent and authoritative. Prior research on clinical decision-support systems shows that automation can improve overall performance while also introducing new errors through over-trust and passive acceptance [4]. In our benchmark, the combination of zero under-triage and moderate over-triage could make an LLM appear “safely conservative,” but this may encourage clinicians to accept high-priority recommendations too readily, even when they are not warranted.

These findings support a **human-in-the-loop** governance model. First, LLM outputs should be treated as advisory rather than final. Second, any deployment should require **model-specific and prompt-specific auditing**, because a prompting strategy that improves one model may worsen another. Third, evaluation should include **subgroup and variant-based fairness testing**, not accuracy alone. Finally, deployment should include clear override mechanisms, logging, and continuous post-deployment monitoring so that emerging bias or instability can be identified early.

5 Conclusion

This project provides a focused benchmark-based case study of LLM-supported emergency triage. Across the tested conditions, DeepSeek and Kimi both showed zero under-triage, indicating a strong safety baseline in this small benchmark. However, both models also showed moderate over-triage, and fairness patterns varied by subgroup, model, and prompt condition. DeepSeek appeared more sensitive to prompt engineering, whereas Kimi was more consistent across demographic variants but could still show subgroup disparities. Overall, our findings suggest that LLMs may be useful as clinical support tools, but not as autonomous triage decision-makers. Their safe use requires structured benchmarking, subgroup auditing, transparency, and sustained human oversight.

References

- [1] Gilboy, N., Tanabe, P., Travers, D., & Rosenau, A. M. (2012). *Emergency Severity Index (ESI): A Triage Tool for Emergency Department Care, Version 4*. Agency for Healthcare Research and Quality.
- [2] Kaboudi, N., Firouzbakht, S., Eftekhar, M. S., & Fayazbakhsh, F. (2024). Diagnostic accuracy of ChatGPT for patients' triage: A systematic review and meta-analysis. *Archives of Academic Emergency Medicine*, 12(1), e60. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11407534/>
- [3] Nedos, I., Zagalioti, S.-C., Kofos, C., Katsikidou, T., Vellidou, D., Astrinakis, K., Karagiannis, I., Giannakopoulos, P., Michaloudi, S., Apostolopoulou, A., Karagiannidis, E., & Fyntanidou, B. (2026). Is Artificial Intelligence Ready for Emergency Department Triage? A Retrospective Evaluation of Multiple Large Language Models in 39,375 Patients at a University Emergency Department. *Journal of Clinical Medicine*, 15(4), 1512. <https://doi.org/10.3390/jcm15041512>
- [4] Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1), 121–127. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3240751/>

Acknowledgements

This project was completed through equal collaboration among all five team members, with each member contributing approximately 20% of the overall work.

- **Wang Sichao** and **Song Xinyi** were responsible for dataset selection, vignette design, and PowerPoint preparation.
- **Li Zhiying** and **Song Siyi** were responsible for implementing the LLM benchmarking pipeline and writing the report.
- **Yang Jiayu** was responsible for data analysis and the initial ethical interpretation.